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188414

Roll No.

1st Sem/ Level - 3 / DVOC
(Ref. & Air Cond., Medical Imaging Tech.,
Auto. Servicing, ITM, PT, SD, AMT, FP, EMS)
Subject : Applied Chemistry

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

Q.1 Electronic configuration of Chromium (Atomic no. =24)

- | | |
|----------------------------|----------------------------|
| a) $[\text{Ar}] 3d^5 4s^1$ | b) $[\text{Ar}] 3d^4 4s^2$ |
| c) $[\text{Ar}] 3d^5 4s^2$ | d) None |

Q.2 NO. of Periods in modern periodic table is

- | | |
|------|-------|
| a) 8 | b) 7 |
| c) 2 | d) 18 |

Q.3 The property of metal to convert it into thin sheets is called

- | | |
|-----------------|----------------|
| a) Ductility | b) Conductance |
| c) Malleability | d) Texture |

Q.4 Shape of p-orbital

- | | |
|-------------------|------------|
| a) Spherical | b) Dumbell |
| c) Double Dumbell | d) None |

Q.5 Permanent Hardness of water is due to

- a) Chlorides of Ca^{2+} & Mg^{2+}
- b) Sulphate of Ca^{2+} & Mg^{2+}
- c) Both
- d) None

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 Give an example of covalent compound.

Q.7 Monomers of Bakelite are ____ and ____.

Q.8 The water which easily forms lather with soap solution is known as ____.

Q.9 Give an example of Liquid fuel.

Q.10 The chemical composition of corrosion product of iron is ____

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

Q.11 Write down any five characteristics of a good fuel

Q.12 Write down any five differences between thermoplastics and thermosetting plastics.

Q.13 Define Covalent bond. What are the properties of covalent compounds?

Q.14 Enlist any five qualities of drinking water.

Q.15 Differentiate between an orbit and orbital.

Q.16 Discuss in brief any two methods of Prevention of Corrosion.

Q.17 Define Corrosion with an example. Write down two effects of Corrosion.

Q.18 Define Ionization Energy. How ionization energy varies in groups and periods?

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 a) Define fuel. Explain any three properties of each CNG and LPG.

b) Explain solid, liquid, gaseous fuels with examples.

Q.20 Write short notes on the following:

a) Isotopes and Isobars

b) Ductility and Malleability